Yaowen Shi

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Education

South China University of Technology (SCUT)
B.S. in Big Data Management and Application *Sep. 2023 – Jun. 2027 (Expected)*

- Academic standing: GPA **3.95/4.00**; average score: **93.7**; major rank: **3/161**
- Selected courses: Calculus I/II (**99, 100**); Linear Algebra and Analytic Geometry (**100**); Operations Research (**99**); Python Data Analysis (**97**); Probability and Statistics (**96**); Large Language Models and GenAI (**96**); Data Structures (**95**); **ranked 1st in 11 courses**.

Research Interests

I work on multimodal agent systems that connect reasoning, planning, memory, and tool use with real creation workflows. My research focuses on LLM-based agents for visual and multimodal generation: agents that understand human intent, decompose open-ended tasks, coordinate specialized tools, and improve outputs through execution feedback. I am broadly interested in image/video/3D generation, visual reasoning, agentic reinforcement learning with executable environments, and autonomous task execution, with the long-term goal of making foundation models more capable and controllable for complex creative tasks.

Publications

SymGround: Visual Program Synthesis with Symbol-to-Vision Grounding
Yaowen Shi, *Jinxiu Liu*
Under Review at EMNLP 2026.

- Formulated visual-to-program credit assignment—grounding rendered residuals in editable program symbols—as a core bottleneck in executable visual synthesis.
- Built a relation-grounded revision loop with visual attribution, evidence-guided editing, and edit-render memory, representing each edit as a testable symbol-to-vision hypothesis.
- Evaluated on SlideBench, BlenderGym, and BlenderBench; outperformed VIGA on BlenderGym (**26.88 → 52.17**) and BlenderBench (**89.12 → 137.99**), cutting invalid executions from **7.47%** to **0.16%** with fewer tokens per task.

Experience

Visiting Student, Westlake University May 2026 – Present
Advised by Yandong Wen, Assistant Professor at Westlake University

- Building a visual "code factory" that automatically constructs executable environments and verifiable tasks for 3D (Blender Python) and web (HTML) code generation, extending SWE-bench-, SWE-bench Pro-, and GDPval-style task construction with minimal human annotation.
- Developing reinforcement learning for visual code agents on top of these environments, using execution- and render-based verification as scalable reward signals.
- Built a benchmark for 3D and HTML code generation, evaluating how agents plan, write, debug, and refine code to produce visual artifacts such as 3D scenes and web pages.

Multimodal Agents for Visual Creation Mar. 2025 – Present
Collaborator: Jinxiu Liu, Research Scientist at Nex-AGI

- Developed SymGround for executable slide and Blender-scene synthesis, separating visual attribution from localized code editing and using edit-render evidence to avoid unsupported retries and broad rewrites.
- Built multimodal generation-agent workflows with NexAU and ComfyUI, defining tool-using agents, skills, and workflow primitives for image/video generation and editing; achieved state-of-the-art results on ComfyBench.
- Explored controllable generation methods including self-reflection/RL for flow-matching text-to-image models, AMF-based video motion transfer with DiT models, and VQ-VAE/adjacency-aware tokenization for 3D meshes.

Projects

OpenDinq | *Evidence-backed AI-native profiles, card workspaces, and explainable people discovery* GitHub

- Built an open-source product alpha that turns public information about a person into reviewable profiles with evidence-backed claims, curated cards, and public profile pages.
- Implemented ingestion from GitHub, websites, OpenAlex, arXiv, ORCID, links, and notes; normalized, deduplicated, and quality-scored claims for explainable people discovery.

Awards

Meritorious Winner (Top 7%), Mathematical Contest in Modeling (MCM) / Interdisciplinary Contest in Modeling (ICM) 2025

Second Prize (Top 15%), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), Guangdong Division 2024

Technical Skills

Languages: C++, Python, SQL, JavaScript, HTML, CSS
ML: PyTorch, Transformers, diffusers
Developer Tools: Git, Docker
English: CET-6